Work-let Area – AI, ML | Controlnet using 2D wireframe/skeleton images

SAMSUNG

Problem Statement

- Goal of project is to create a controlnet based diffusion model using 2D wireframe/skeleton images as control
- Current skeleton based controlnet supports only human and hand image generation
- The goal of this project is to generate 2D images of any object using a text prompt and a skeleton/wireframe images as control
- The scope of the project includes collection of dataset and training a large diffusion model

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Additional Documentation:

- https://objaverse.allenai.org/
- https://huggingface.co/blog/train-your-controlnet

Expectations

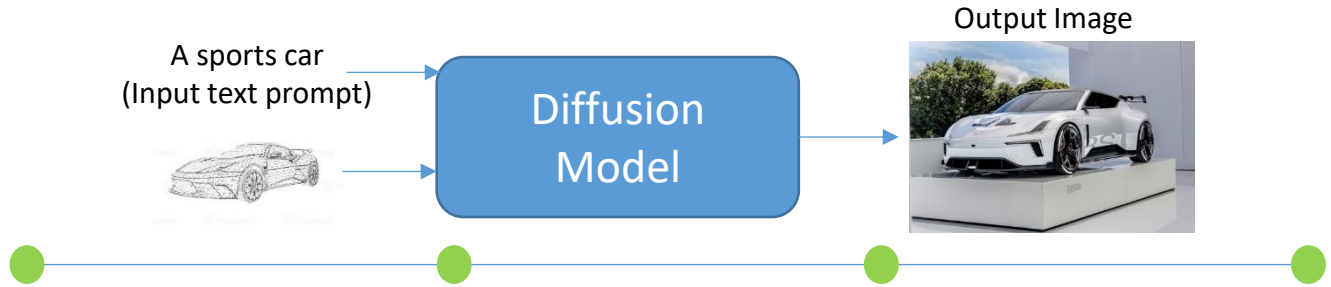
Work-let expected duration – 6 months

3 Students

- Explore objaverse dataset and controlnet finetuning
- Develop a method to create 3D wireframe/skeleton from objaverse dataset along with text prompts to create training set
- Render 2D images from the objaverse dataset and associated wireframe/skeleton images for control
- Train a controlnet using the text and skeleton/wireframe as control and 2D images as output

Training/ Pre-requisites

- Knowledge of diffusion models and 3D graphics
- Hands on in Deep learning development frameworks such as pytorch and huggingface
- GPUs (A100 – 80 GB or better)



Kick Off < 1st Month >

- Understanding diffusion models and finetuning diffusion models
- Explore controlnet and controlnet training
- Explore objaverse dataset

Milestone 1 < 2nd Month >

- Create 3D skeleton/wireframes from objaverse dataset
- Feasibility study and choose either wireframe/skeleton for the remaining duration
- Start training controlnet

Milestone 2 < 4th Month >

- Training controlnet and improvements

Closure < 6th Month >

- Provide the final controlnet model that can create high fidelity 2D images using a text prompt and skeleton/wireframe as contol

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What I went through

- https://github.com/lllyasviel/ControlNet
- https://github.com/kennymckormick/pyskl
- https://github.com/CMU-Perceptual-Computing-Lab/openpose
- https://objaverse.allenai.org/
- https://huggingface.co/blog/train-your-controlnet